

Relative price dispersion and inflation regimes in South Africa

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Motivation

Motivation

- ▶ Inflation is known to have adverse welfare effects through its distortion of relative prices which causes the misallocation of resources, production inefficiencies leading to lower real economic activity.
- ▶ This is well supported by both theoretical and empirical studies.
- ▶ This paper contributes to the empirical literature, by examining the relationship between inflation and relative price variability (RPD) in South Africa given the little evidence available and to better understand how this relationship has evolved and the implications for monetary policy.

Motivation

- ▶ The South African context can provide further insights into the relationship between inflation and RPD given the shifts in the inflation targets from 3-6 per cent in 2000 to 4.5 per cent since 2017 as structural changes in inflation can affect the link between inflation and relative price variability (Caglayan and Filiztekin 2003)
- ▶ That the functional form of relationship is U-shaped and dynamic (Fielding and Mizen 2000; Choi 2010)
- ▶ The latter motivates the estimation of the inflation rate that minimizes relative price distortions since if the relationship was linear then the optimal inflation rate would be zero
- ▶ Given the inflation target reform towards a 3% in South Africa, this study contributes to the policy discussion on the appropriate inflation target and understating how dynamic relations between inflation and relative price variability has evolved

Theoretical Framework: RPD through Lucas's
aggregate supply curve

RPD theoretical framework

- ▶ The theoretical literature generally distinguishes between menu-cost and signal extraction models in explaining the positive relation between trend inflation and RPD.
- ▶ In these models' firms follow a one-sided dynamic pricing rule when setting prices and face fixed non-zero costs of adjusting nominal prices (e.g. printing new menus) when inflation increases as the latter reduces the optimal real price of goods and services.
- ▶ Due to differences in administrative costs, these nominal price adjustment costs vary across firms moreover these adjustments occur at different times which implies staggered price setting behaviour which leads to relative price variability.

RPD theoretical framework

- ▶ Menu cost models predict a that higher expected inflation causes an increase in RPD (citations xxx) while Lucas-type signal extraction models (Lucas 1973; Hercowitz 1981; Cukierman 1983) argue unexpected inflation uncertainty causes higher RPD because informational frictions make it difficult to distinguish between idiosyncratic relative shocks or an aggregate (inflationary) shock.
- ▶ Menu cost and signal-extraction models assume a positive and linear relationship making the optimal inflation rate zero, however, recently monetary search models (Head and Kumar 2005) show that this relationship is nonlinear making the optimal inflation rate somewhere between zero and positive inflation rate.

RPD theoretical framework

- ▶ The concept of relative price dispersion can be abstract and difficult to grasp without a theoretical framework.
- ▶ One of the most influential frameworks for understanding relative price dispersion is the Lucas (1973) aggregate supply curve, which provides insights into how relative price dispersion can explain the output-inflation relationship.

RPD theoretical framework

- ▶ This theory essentially means (in part) that nominal “rigidities” partially dominate short-run supplier behaviour resulting from a lack of information about the some prices relevant to decision making.
- ▶ For our purposes we ignore other aspects of the theoretical framework and emphasis this aspect of the model to provide an understanding relative price dispersion.

RPD theoretical framework

Lucas (1973) proposes a decomposition of the supply curve, all in logs, into a natural or secular component and a cyclical component as follows:

$$y_t(z) = y_{n,t} + y_{ct}(z), \quad (1)$$

- ▶ $y_t(z)$ is the aggregate supply at time t in market z , $y_{n,t}$ is the natural or secular level of supply at time t
- ▶ y_{ct} is the cyclical component of supply at time t .
- ▶ Competitive markets and a cleared labour market are assumed
- ▶ Leading to relative price and general price level movements leading.
- ▶ Imperfect information of the suppliers about the general price level that leads to relative price dispersion and the output-inflation relationship.

RPD theoretical framework

In addition, the natural or secular level of supply is assumed to follow a linear trend as follows:

$$y_{n,t} = \alpha + \beta t, \quad (2)$$

where α is a constant and β captures the trend growth rate of the natural level of supply.

RPD theoretical framework

The cyclical component varies with the perceived relative price (the difference between the actual price level and the expected price level) and its lags as follows:

$$y_{ct}(z) = \gamma[P_t(z) - E(P_t|I_t(z))] + \lambda y_{c,t-1}(z), \quad (3)$$

- ▶ γ captures the sensitivity of supply to the perceived relative price, $P_t(z)$ is the actual price level in market z at time t ,
- ▶ $E(P_t|I_t(z))$ is the expected price level based on information available at time t in market z
- ▶ λ captures the persistence of the cyclical component and $|\lambda| < 1$. $E(P_t|I_t(z))$ is current expected price level conditioned on information that is available at time t , at market z .
- ▶ The available information set ($I_t(z)$) comprises of past demand driven shifts of the secular component ($y_{n,t}$),
- ▶ Past deviations of the cyclical component ($y_{c,t-1}, y_{c,t-2}, \dots$) from its natural level.

RPD theoretical framework

- ▶ The current general price level (P_t), which determines 'prior' expectations common across all suppliers in all markets, is assumed to be $P_t \sim N(\overline{P}_t, \sigma^2)$
- ▶ $\overline{P}_t$ is the mean price level at time t and σ^2 is the variance of the price level at time t .
- ▶ Furthermore, Lucas (1973) defines z_t as the percentage deviation of the actual market price from the general price level as $z \stackrel{\text{iid}}{\sim} N(0, \tau^2)$, where τ^2 is the variance of z .

RPD theoretical framework

Therefore, the actual price level in market z at time t can be expressed as:

$$P_t(z) = P_t + z, \quad (4)$$

- ▶ $P_t(z)$ is the actual price in market z at time t (that is the markets are indexed to z),
- ▶ P_t is the current general price level at time t , and z is the percentage deviation of the actual market price from the general price level.
- ▶ However, P_t is unobserved by traders, and therefore $I_t(z)$ is used to estimate the expected price level ($E(P_t|I_t(z))$).

RPD theoretical framework

The expected price level based on information available at time t in market z can be derived using Bayes' rule as follows:

$$E(P_t|I_t(z)) = E(P_t|P_t(z), \overline{P}_t) = (1 - \theta)P_t(z) + \theta\overline{P}_t, \quad (5)$$

- $\theta = \frac{\tau^2}{(\sigma^2 + \tau^2)}$ is the weight placed on the mean price level ($\overline{P}_t$) in forming expectations.

RPD theoretical framework

Combining Equation ??, Equation ??, and Equation ?? by substitution yields the following:

$$y_t(z) = y_{nt} + \theta\gamma[P_t(z) - \overline{P_t}] + \lambda y_{c,t-1} \quad (6)$$

- current supply in market z at time t is a function of the natural level of supply, the perceived relative price (the difference between the actual price level in market z and the mean price level), and the lagged cyclical component.

RPD theoretical framework

Finally, aggregating across all markets (or integrating over all markets) yields the following aggregate supply curve:

$$y_t = y_{nt} + \theta\gamma(P_t - \overline{P}_t) + \lambda[y_{t-1} - y_{n,t-1}] \quad (7)$$

From Equation ??, we can clearly demonstrate how the relative price dispersion (θ) is a fraction of the total individual price dispersion ($\sigma^2 + \tau^2$), affects the aggregate supply curve, and therefore the output-inflation relationship.

RPD theoretical framework

- ▶ That is, $\theta \uparrow$ in τ^2 and $\downarrow$ in σ^2 .
- ▶ Meaning that when $\theta \uparrow$, the aggregate supply curve becomes flatter, and the output-inflation relationship becomes stronger.
- ▶ And when $\theta \downarrow$, the slope is restricted by γ .
- ▶ Therefore, the relative price variance (θ), in this context, is a key determinant of the slope of the aggregate supply curve and the strength of the output-inflation relationship.

Empirical literaturer review

Data and methodology

Data

- ▶ Our analysis uses monthly inflation data on 45 goods categories from January 2009 to November 2025 obtained from Statistics South Africa (Stats SA).¹
- ▶ Figure ?? below shows the long term inflation rate in South Africa, and Figure ?? in the appendix shows the various price categories used in the analysis

¹The data is available at: <http://www.statssa.gov.za/> and the data was seasonally adjusted.

Data

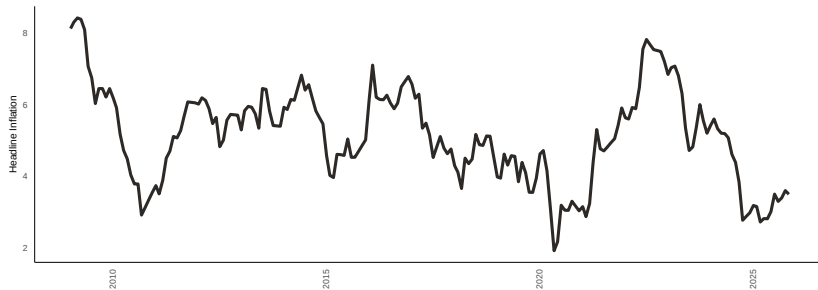


Figure 1: Headline inflation (%).

Data

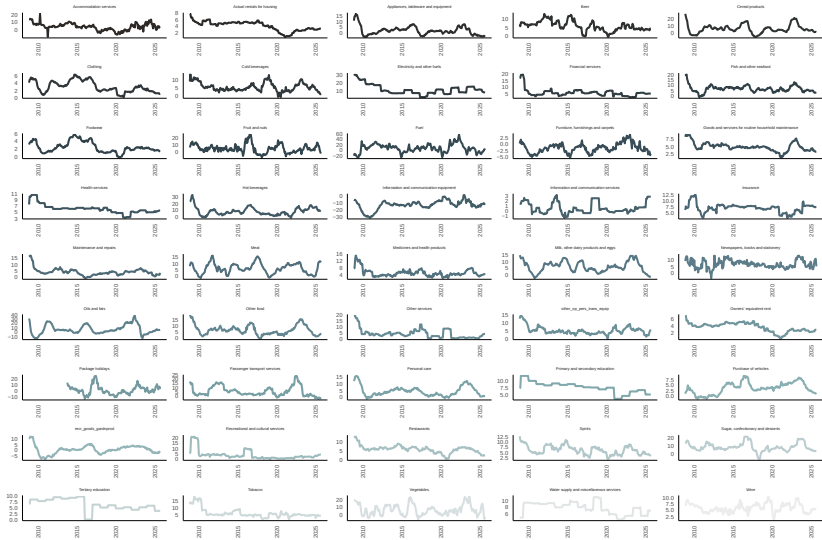


Figure 2: Inflation categories (%)

Methodology

RPD is calculated in line with Lach and Tsiddon (1992) as the weighted standard deviation of individual price inflation rates from the aggregate inflation rate. The RPD is essentially the variability of relative prices of a given product. - We first calculate the monthly inflation rate as follows:

$$\pi_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right) * 100, \quad (8)$$

- ▶ $\pi_{i,t}$ is the inflation rate of good i at time t
- ▶ $P_{i,t}$ is the price of good i at time t
- ▶ $P_{i,t-1}$ is the price of good i at time $t - 1$

Methodology

Then, the RPD is calculated as:

$$RPD_t = \sqrt{\sum_{i=1}^n \omega_i (\pi_{i,t} - \pi_t)^2}, \quad (9)$$

- ▶ ω_j is the expenditure share of the j^{th} good,
- ▶ π_t is the headline or aggregate inflation rate at time t ,
- ▶ $\sum_{i=1}^n \omega_i = 1$.
- ▶ The expenditure weights (ω_i) used are derived from the Consumer Price Index (CPI) basket provided by Stats SA

Methodology

To examine the relationship between inflation and RPD, we estimate the following econometric model:

$$RPD_t = \beta_0 + \beta_1 |\pi_t| + \sum_{h=1}^3 \omega_h RPD_{t-h} + \mu_t, \quad (10)$$

- ▶ RPD_t is the relative price variance at time t
- ▶ $|\pi_t|$ is the absolute value of aggregate inflation rate at time t
- ▶ RPD_{t-h} are lagged values of RPD to account for its dynamic nature, and μ_t is the error term
- ▶ The coefficient β_1 captures the effect of inflation on RPD. We include three lags of RPD based on Akaike Information Criterion (AIC) (and other tests) and to capture potential persistence in RPD.
- ▶ For robustness, we repeat this estimation using core inflation (excluding food and non-alcoholic beverages, and fuel and electricity) instead of headline inflation to account for the potential volatility in these components

Methodology

The literature on RPD emphasises that the relationship between inflation and RPD maybe time-varying due to structural changes in the economy, monetary policy regimes, and other factors. To capture potential time variation in the relationship, we also estimate a rolling regression model with a 24-month and 60-month windows as follows:

$$RPD_t = \alpha_t + \beta_{1,t}|\pi_t| + \sum_{h=1}^3 \omega_{h,t}RPD_{t-h} + \mu_t, \quad (11)$$

- ▶ The coefficients α_t , $\beta_{1,t}$, and $\omega_{h,t}$ vary over time.
- ▶ This approach allows us to observe the stability of the coefficients over time, which is an indicator of non-linearity in the relationship between RPD and inflation.

Methodology

Next, we utilise a semi-parametric or partially linear approach to estimate the optimal level of inflation given the RPD. This approach allows us to model the relationship between inflation and RPD without imposing a strict functional form. Following the literature, we first specify a quadratic (U-shaped) relationship between inflation and RPD as follows:

$$RPD_t = \alpha + \pi_t + \beta_2 \pi_t^2 + \sum_{h=1}^3 \phi_h RPD_{t-h} + \epsilon_t, \quad (12)$$

- ▶ β_2 captures the curvature of the relationship between inflation and RPD
- ▶ The optimal inflation rate that minimizes RPD can be derived by taking the first derivative of the equation with respect to π_t and setting it to zero, yielding:

$$\pi_t^* = -\frac{1}{2\beta_2}. \quad (13)$$

Methodology

To estimate this optimal inflation we implement a semi-parametric approach as follows:

$$RPD_t = f(\pi_t) + \sum_{h=1}^3 \phi_h RPD_{t-h} + \epsilon_t, \quad (14)$$

- ▶ $f(\pi_t)$ is a non-linear smooth differentiable function of inflation and the RPD.
- ▶ Therefore, the optimal inflation rate can be obtained by finding the minimum of the estimated $f(\pi_t)$ function. $f(\pi_t)$ is estimated in two stages.

Methodology

First we estimate λ_h as follows:

$$RPD_t = \sum_{h=1}^3 \lambda_h \overline{RPD}_{t-h} + \eta_t, \quad (15)$$

- $\overline{RPD}_{t-h}$ are the residuals from a non-parametric regression of each lag of RPD_t on π_t

Methodology

In the second stage, we estimate $f(\pi_t)$ non-parametrically as follows:

$$\hat{\eta}_t = f(\pi_t) + v_t, \quad (16)$$

- ▶ $\hat{\eta}_t$ are the residuals from Equation ??
- ▶ $f(\pi_t)$ is estimated using a kernel regression method
- ▶ The bandwidth (h) for the kernel estimator is selected using cross-validation to minimize the mean squared error.

Methodology

- ▶ As a treatment for extreme values of inflation, we further implement the unbounded Gaussian kernel and the outlier-robust Epanechnikov kernel.
- ▶ In essence, the semi-parametric approach captures the sensitivity of RPD to marginal increases in inflation.
- ▶ In that case, the optimal inflation rate can be obtained by finding the point where the first derivative ($f'(\pi_t)$) is equal to zero.
- ▶ Therefore, if $f'(\pi_t) > 0$ ($f'(\pi_t) < 0$), then an increase in inflation (decrease) leads to an increase (decrease) in the RPD. For robustness we estimate the $g(UIN_t)$ with multiple bandwidth (h).

Methodology

To further disentangle the effects of RPD on inflation, we decompose headline inflation into its expected and unexpected components. As discussed in the literature section, for example, theory suggests that inflation uncertainty has a more pronounced effect on RPD due to information frictions faced by firms when adjusting prices (Lucas 1973; Hercowitz 1981; Cukierman 1983).

We therefore estimate the following model:

$$RPD_t = \alpha + \beta_0 EIN_t + \beta_1 UIN_t + g(UIN_t) + \sum_{k=1}^3 \lambda_k RPD_{t-k} + \epsilon_t, \quad (17)$$

- ▶ EIN_t is the expected inflation rate and UIN_t is the unexpected inflation rate
- ▶ $g(UIN_t)$ is a non-linear function of unexpected inflation (UIN_t).
- ▶ We obtain the expected using **a simple moving average of**

Methodology

Lastly, and some-what novel to the RPD literature, we employ a quantile on quantile regression approach (Koenker 2017). Quantile on quantile regressions are robust to outliers and allows us to estimate the regime aspects of the relationship between RPD and inflation.

For brevity, we do not explain the quantile on quantile estimation procedure

Results

Results

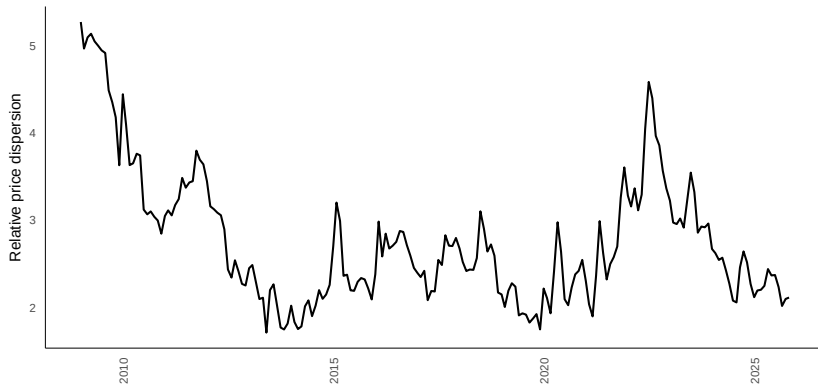


Figure 3: RPD over-time

Results

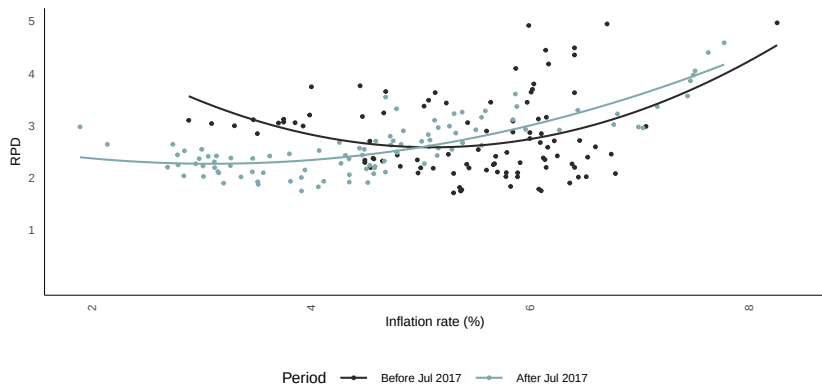


Figure 4: RPD vs headline Inflation